Neural Network models (Time-wise split): Metrics, Voltage and Power plots

Here, we perform time-wise split to capture payload status as well in the test data. Roughly, due to constraints and to prevent data leakage we are dividing data on basis of time and the approximate split is Train=52% and Test=48%

Here, we use 3 neural network models for voltage and power comparison:

1. Dense Neural Network (DNN)
2. Deep Neural Network (DeepNN)
3. Long Short Term Memory (LSTM)

Following are the metrics:

Voltage Prediction:

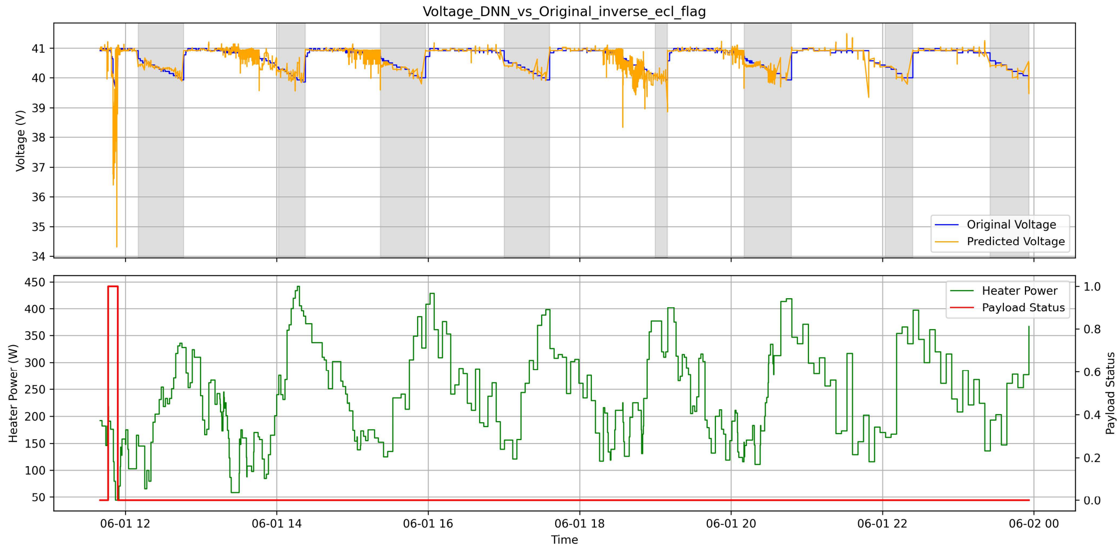
Model	MAE (V)	RMSE (V)	R^2
DNN	0.1124	0.2402	0.353
DeepNN	0.1526	0.5023	-1.828
LSTM	0.0626	0.0903	0.909

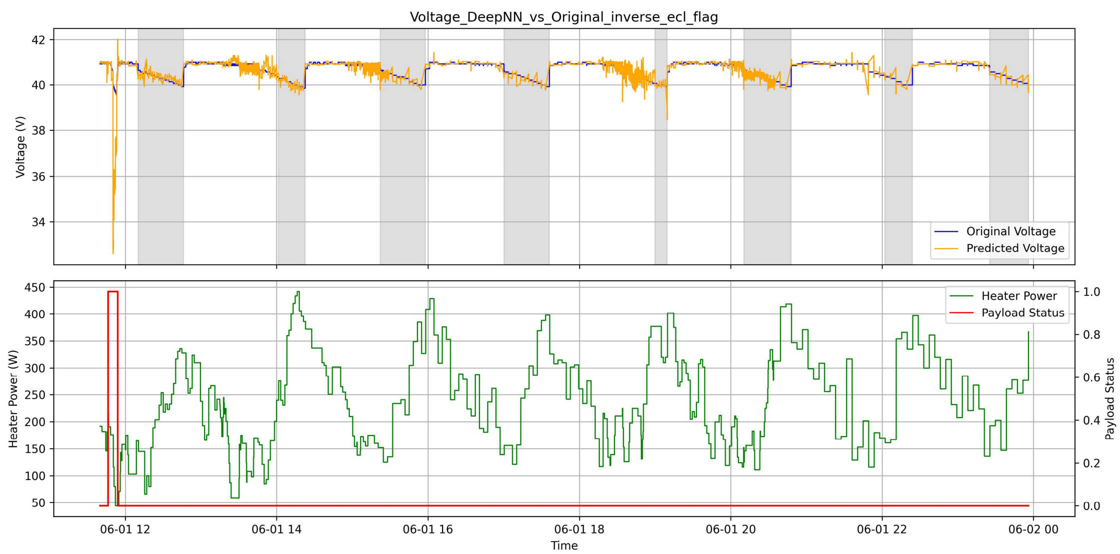
Power Prediction:

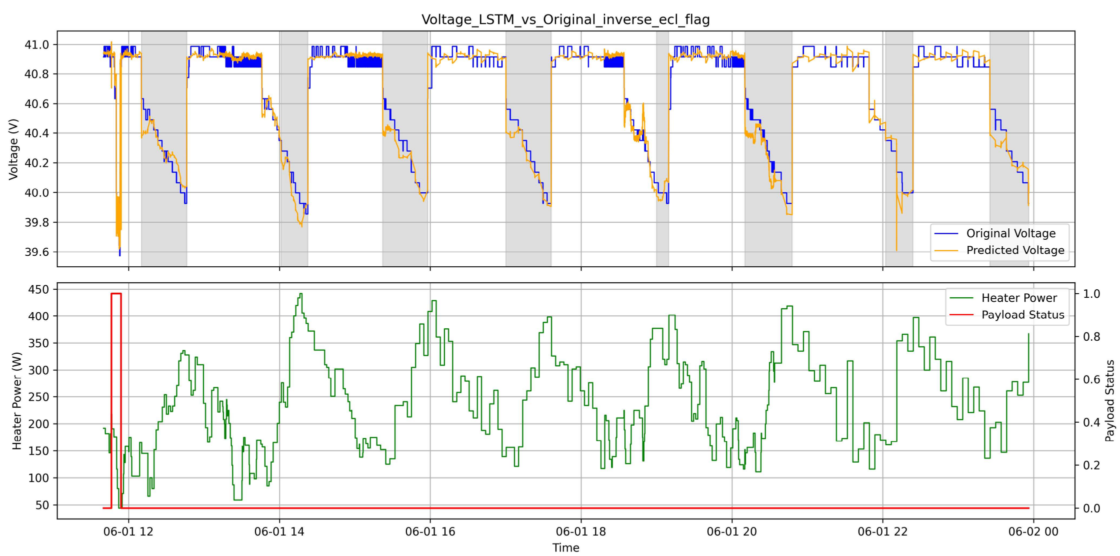
Model	MAE (W)	RMSE (W)	R^2
DNN	3.3568	13.1208	0.9919
DeepNN	2.2117	6.1792	0.9982
LSTM	13.8612	71.9230	0.756

- **LSTM performs best for voltage due to temporal dynamics**
- **DeepNN performs best for power due to strong instantaneous feature relationships**

Voltage Graph Plots:







Power graph plots: